Semester End Examination Solutions

Problem Solving and Computer Programming (E1PA103B)

Institution: Galgotias University
Course: Masters of Computer Applications (MCA)
Duration: 120 Minutes | **Max Marks:** 50

1. Format Specifiers in C

Format specifiers define the type of data to be displayed or read. [cite: 1]

Specifier	Data Type	Example
%d	Signed Integer	`printf("%d", 10);`
%f	Float	`printf("%f", 3.14);`
%c	Character	`printf("%c", 'A');`
%s	String	`printf("%s", "Hello");`

2. Compilation and Execution Process

The steps involved in the compilation and execution of a C program are: [cite: 1]

- **Preprocessing:** Expanding macros and processing header files.
- **Compilation:** Converting preprocessed source code to assembly code.
- **Assembly:** Converting assembly code into machine-readable object code.
- **Linking:** Combining object files with libraries to create an executable.

3. Arrays as Pointers

In C, an array name acts as a constant pointer to the first element of the array. [cite: 1]

```
int arr[3] = {10, 20, 30};
// 'arr' holds the address of arr[0]
printf("%d", *arr); // Outputs 10
```

4. Structure vs. Union

Comparison between Structure and Union based on memory management: [cite: 1]

Feature	Structure	Union
Memory	Unique storage for each member.	Shared storage for all members.
Size	Sum of all members.	Size of the largest member.
Access	All members simultaneously.	One member at a time.

5. 2D Array Row Sum Program

C program to calculate the sum of each row in a 3x3 matrix: [cite: 1]

```
#include <stdio.h>
int main() {
    int mat[3][3], i, j, sum;
    for(i=0; i<3; i++)
        for(j=0; j<3; j++)
            scanf("%d", &mat[i][j]);

    for(i=0; i<3; i++) {
        sum = 0;
        for(j=0; j<3; j++) sum += mat[i][j];
        printf("Row %d Sum: %d\n", i+1, sum);
    }
    return 0;
}
```

6. Partial Initialization

If an array is declared as `int numbers[5] = {10, 20};`, the behavior is: [cite: 1]

- `numbers[2]` = 0
- `numbers[3]` = 0
- `numbers[4]` = 0

7. Pointers and Preprocessors

(a) Null Pointer: A pointer initialized to point to nothing using the keyword `NULL`. [cite: 1]

(b) Pointer Operation:

```
int num = 100;
int *ptr = &num;
*ptr += 50;
printf("%d", num); // Output: 150
```

(c) Preprocessors: `<file.h>` searches system paths, while `"file.h"` searches the current directory. [cite: 1]